

Sheikh Umer

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SUMMARY

Principal Data Architect with senior-level expertise delivering scalable, AI-ready data platforms for real-time banking analytics and risk modeling. Led end-to-end cloud migration and built modular pipeline frameworks on AWS using Spark, Kafka, Airflow, and Terraform, improving data reliability and regulatory compliance. Previously designed a modern healthcare data platform ingesting clinical and genomic data, enabling secure, HIPAA-compliant analytics across Snowflake and Redshift. Seeking to apply deep data engineering and architecture skills to accelerate data-driven decision making and operational excellence.

SKILLS

- **Data Engineering:** Apache Airflow, dbt, Dagster, Prefect, AWS Glue, Azure Data Factory
- **MLOps & Machine Learning:** MLflow, Kubeflow, Vertex AI, SageMaker
- **Cloud Platforms:** AWS, S3, Redshift, Glue, Athena, Lambda, RDS, EMR, CloudFormation, GCP, BigQuery, Dataflow, Vertex AI, Composer, Azure, Data Factory, Synapse, Event Hubs, Data Lake
- **Programming & Scripting:** Python, Pandas, PySpark, FastAPI, SQL, T-SQL, PL/pgSQL, Scala, Java, Bash, Go
- **Analytics & Data Warehousing:** Snowflake, BigQuery, Amazon Redshift, Azure Synapse
- **Real-Time & Messaging Systems:** Apache Kafka, Kafka Streams, Kinesis, Flink CEP, Debezium, CDC, Pub/Sub, Apache Pulsar
- **DevOps & Infrastructure as Code:** Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, GitLab CI/CD, ArgoCD
- **Distributed Processing & Big Data:** Apache Spark, Apache Flink, Databricks, Dask, Ray, EMR, Hadoop, Beam, Delta Lake, Iceberg, Hudi, Parquet, ORC
- **Security & Compliance:** IAM, RBAC/ABAC, HashiCorp Vault, Apache Atlas, DataHub, Colibra, GDPR, HIPAA, SOC 2, ISO 27001
- **Big Data & Distributed Processing:** Apache Spark, Apache Flink, Databricks, Hadoop, Apache Beam, EMR
- **Monitoring & Observability:** Prometheus, Grafana, Datadog, CloudWatch, ELK Stack, OpenTelemetry, New Relic
- **Visualization:** Looker, Power BI, Tableau, Superset, Sigma
- **Compliance & Regulatory:** HIPAA, GDPR, SOC 2, ISO 27001
- **Streaming & Messaging:** Apache Kafka, Kafka Streams, Amazon Kinesis, Google Pub/Sub, Debezium, CDC, Apache Pulsar
- **Data Lake & Storage Formats:** Delta Lake, Apache Iceberg, Apache Hudi, Parquet, ORC
- **Data Warehousing & Analytics:** Snowflake, BigQuery, Amazon Redshift, Azure Synapse

PROFESSIONAL EXPERIENCE

BMC

Mar 2023 - Present

Principal Data Architect

- Architected AI-ready data platforms with MLflow and Apache Atlas that support real-time banking analytics, risk modeling, and fraud detection, enabling faster insight generation for the finance team
- Designed a scalable cloud data architecture on AWS (Glue, Athena, Redshift) with microservices-based APIs, which streamlined secure financial data integration and cut data onboarding time
- Implemented version-controlled data models and data contracts using Git, improving collaboration among engineering, analytics, and compliance teams and accelerating model delivery
- Built modular data pipeline frameworks and reusable Terraform modules, standardizing infrastructure-as-code deployments and reducing deployment effort
- Advised leadership on data governance, privacy, and regulatory compliance for scalable financial data systems, helping the organization meet GDPR requirements and avoid compliance penalties
- Implemented observability using Prometheus, OpenTelemetry, and Grafana to strengthen monitoring and operational reliability.
- Supported SOC 2, PCI DSS, and GDPR compliance initiatives for regulated financial environments by designing data-lineage tracking in Apache Atlas and enforcing encryption policies, ensuring audit readiness and regulatory adherence

3Cloud

Aug 2020 - Feb 2023

Data Platform Architect

- Designed and led the implementation of a modern data platform on AWS using Redshift, Dask, and Docker to ingest, store, transform, and serve clinical, genomic, and real-world healthcare data, which streamlined data access for researchers and reduced onboarding time
- Architected data lakes, lakehouses, and warehouses on AWS S3, Redshift, Snowflake, and Delta Lake, establishing a unified storage layer that improved query performance and simplified data governance
- Defined standards for data ingestion pipelines, metadata management, data quality validation, and API-based access layers using AWS Glue and custom Dask workflows, resulting in consistent data quality and faster downstream analytics
- Collaborated with engineering and DevOps teams to implement Infrastructure as Code with Terraform, CI/CD pipelines via ArgoCD, and automated testing frameworks, which accelerated deployment cycles and ensured reliable infrastructure provisioning
- Enabled batch, micro-batch, and streaming ingestion pipelines integrating data from EHR systems, labs, wearables, and external providers using ArgoCD and Dask, delivering near-real-time data availability for clinical decision support

- Ensured platform security, governance, and HIPAA/HITRUST compliance by configuring Redshift encryption, deploying Docker containers with secure settings, and using ArgoCD for automated policy enforcement, resulting in zero compliance findings during audits
- Implemented data lineage and cataloging using AWS Glue and Dask, collaborating with data-governance teams to define policies, which improved data discoverability and reduced time to locate datasets
- Led architecture reviews and performance tuning of Redshift clusters and Dockerized services, applying cost-optimization strategies with ArgoCD, which lowered monthly cloud spend and improved query response times
- Drove adoption of FHIR and HL7 standards across the platform, integrating them into Redshift schemas and Dask pipelines, enabling healthcare analytics and AI models to access standardized data, which accelerated project delivery

Cruise

Jun 2019 - Jul 2020

Distributed Systems & Cloud Data Pipelines

- Built Python ETL pipelines that extracted marketing data from Pub/Sub, transformed it with Databricks, and loaded it into reporting tables, enabling SMB clients to generate analytics dashboards faster
- Orchestrated data workflows with Apache Airflow and scheduled batch jobs in AWS Glue, which reduced pipeline failures and ensured daily data availability
- Maintained the Amazon Redshift warehouse, applying distribution and sort keys to tables, which improved query performance and lowered storage costs
- Developed metadata-driven ingestion templates, reducing onboarding time for new data sources.
- Created an internal tool that turned ad-hoc SQL queries into reusable dbt models, cutting development time for new transformations and improving code consistency

EDUCATION

University Of Punjab

Bachelor of Science